The Alternate Segment Theorem

Exam code: 0607

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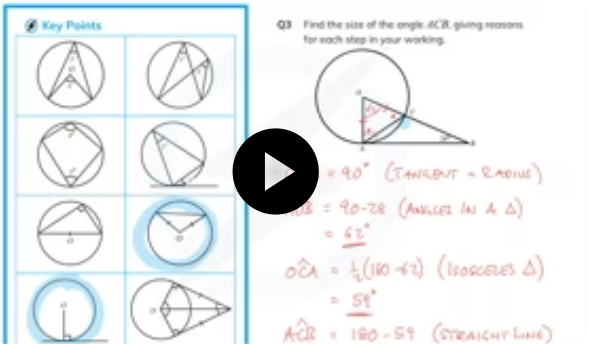
Updated on 23 January 2026



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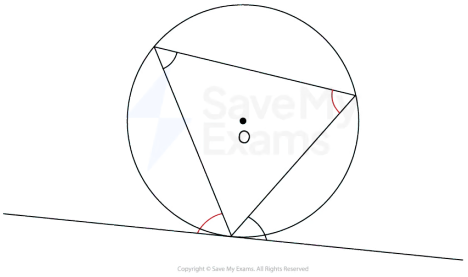
Alternate segment theorem

Circle theorem: The alternate segment theorem

- The **angle** between a **chord** and a **tangent** is **equal** to the **angle** in the **alternate segment**
  - The alternate segment is the region on the **opposite side** of a chord from a given angle formed between that chord and a tangent line



Example of angles in alternate segments

- To spot this circle theorem on a diagram
  - look for a **cyclic triangle**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- To identify which angles are equal
  - mark the **angle** between the **tangent** and the **side** of the **cyclic triangle**
  - the angle **inside** the triangle at the corner **opposite the side** of the triangle that forms the first angle is the **equal angle**
- When explaining this theorem in an exam you can just say the phrase:
  - **The Alternate segment theorem**

Examiner Tips and Tricks

Look for **cyclic triangles** and **tangents** in busy diagrams.

Questions involving the alternate segment theorem frequently appear in exams!

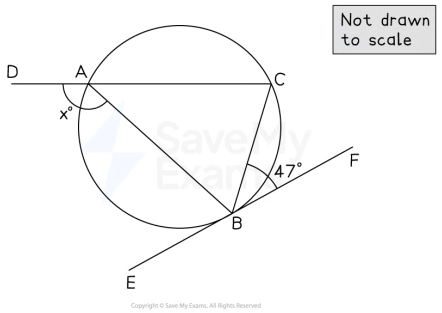
Worked Example

A , B and C are points on a circle.

DAC is a straight line.

EBF is a tangent to the circle.

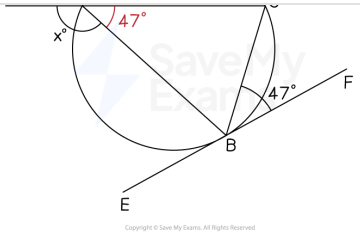
Find the value of x .



Answer:

One vertex of this triangle meets a tangent at point *B*
The angle between one of its sides (*BC*) and the tangent is given
Find the angle inside the triangle, opposite to the same side (*BC*)

Angle *CBF* = Angle *CAB* by the alternate segment theorem



Angle x and angle CAB form a straight lie

$$x + 47 = 180$$
$$x = 180 - 47$$
$$x = 133$$

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